kSZ^2×LAE cross-correlation: signal-to-noise pipeline, validation against La Plante et al. (2022), and detectability for current and future CMB and LAE surveys

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Summary

- We forecast the signal-to-noise (S/N) of the cross-correlation between the filtered and squared CMB temperature map (the kSZ^2 map) and Lyman-α emitters (LAEs), with the estimator of La Plante+22 (their Eq. 13). The CMB noise, the LAE shot noise and the sample (cosmic) variance are all included (Sec. 1).
- The signal comes from our simulation (21cmFAST v4, ten 300 Mpc boxes): either La Plante's galaxy model (matter × Waters+16 bias) or the simulated LAEs, rescaled to each survey's bias (Sec. 2.3).
- **Validation** (Sec. 3). On La Plante's exact setup our noise matches theirs bin by bin and our galaxy clustering matches their Fig. 8. Our total S/N is 0.7–0.85 of their Table 2 with shot noise. The gap comes from the modes our box does not contain ($\ell \lesssim 240$) and from a signal about two times lower at $\ell \gtrsim 1000$ (not yet explained).
- **Detectability** (Sec. 4). None of the requested CMB×LAE pairs reaches 1σ with the validated signal model. Existing data (ACT DR6 × SILVERRUSH) give ∼0.01. A better CMB alone (SO LAT) does not help, because the LAE fields are only 10–20 deg^2. CMB-HD × a 1 deg^2 Roman deep field reaches 0.04–0.2 (0.2–1.2 with the simulated-LAE signal, whose excess is not yet confirmed). A 3σ detection with CMB-HD needs Roman-deep-quality LAEs over ∼200–7000 deg^2.
- **Wider Roman field.** Maitra et al. (2025) list ∼16 deg^2 for the Roman grism. At the depth of the 1 deg^2 deep field this gives 0.4 (1.0 with white noise) for CMB-HD with the validated signal model; this is an upper estimate, since no Roman survey of that size reaches that depth. The real wide-ish tier (HLWAS deep tier, 19.2 deg^2) has a shallower grism and gives ∼0.04.

1 Estimator

S/N. Following La Plante+22 (Eq. 13; Gaussian approximation, after Ferraro et al. 2016), for one galaxy redshift window

$$\left(\frac{S}{N}\right)^2 = f_{\rm sky} \sum_{\ell=\ell_{\rm min}}^{\ell_{\rm max}} \frac{(2\ell+1)\left(C_\ell^{\mathrm{kSZ}^2\times g}\right)^2}{C_f^{T^2T^2}\, C_\ell^{gg} + \left(C_\ell^{\mathrm{kSZ}^2\times g}\right)^2}, \qquad \ell_{\rm max} = 10^4. \tag{1}$$

The three noise sources enter as follows: the CMB noise through $C_f^{T^2T^2}$; the LAE shot noise through the Poisson part of C_ℓ^{gg}; and the sample variance through the clustering part of C_ℓ^{gg}, the $(C_\ell^{\mathrm{kSZ}^2\times g})^2$ term, and the finite number of modes $f_{\rm sky}(2\ell+1)$. The spectra are binned in ℓ, so the sum over a bin uses $\sum_{\ell\in\mathrm{bin}}(2\ell+1) \simeq \ell_{\rm hi}^2 - \ell_{\rm lo}^2$. Several redshift shells of one survey are combined in quadrature.

CMB filter (La Plante+22 Eqs. 8, 10):

$$f(\ell) \propto F(\ell)\, b(\ell), \qquad F(\ell) = \frac{C_\ell^{\rm kSZ,reion}}{C_\ell^{TT} + C_\ell^{\rm kSZ,reion} + C_\ell^{\rm kSZ,late} + N_\ell}, \qquad b(\ell) = e^{-\theta^2\ell^2/(16\ln 2)}, \tag{2}$$

with θ the beam FWHM. f is normalised to 1 at its peak (as in La Plante+22 Fig. 2); the S/N does not depend on this normalisation because signal and noise use the same f.

Noise of the filtered, squared map (La Plante+22 Eqs. 14–15):

$$C_f^{T^2T^2} = 2\int \frac{\ell\,\mathrm{d}\ell}{2\pi}\Big[C_\ell^{\bar T\bar T,f}\Big]^2, \qquad C_\ell^{\bar T\bar T,f} = f(\ell)^2\big(C_\ell^{TT} + C_\ell^{\rm kSZ,reion} + C_\ell^{\rm kSZ,late} + N_\ell\big). \tag{3}$$

CMB noise N_ℓ. Two cases:

$$\text{instrument only (La Plante+22 Eq. 11):} \quad N_\ell = \Delta_T^2\, e^{\theta^2\ell^2/(8\ln 2)}, \tag{4}$$

$$\text{post-ILC:} \quad N_\ell = N_\ell^{\rm ILC} \quad \text{(La Plante+22 Fig. 1).} \tag{5}$$

$N_\ell^{\rm ILC}$ is the instrument noise plus the residual foregrounds after the internal linear combination of the frequency channels. Post-ILC curves exist only for SO, CMB-S4 and CMB-HD. ACT DR6 and SPT-3G are therefore used with instrument noise only, which is optimistic.

Galaxy auto-power (La Plante+22 Eq. 16):

$$C_\ell^{gg} = C_\ell^{\rm clust} + C^{\rm SN}, \qquad C^{\rm SN} = \frac{1}{\bar n_{2D}} = \frac{4\pi f_{\rm sky}}{N_g}, \tag{6}$$

with $\bar n_{2D}$ the number of LAEs per steradian in the shell.

LAE number density. From a Schechter fit to the observed Lyα luminosity function (LF), per dex,

$$n_{3D}(>L_{\rm min}) = \int_{\log L_{\rm min}} \ln 10\, \phi^*\Big(\frac{L}{L^*}\Big)^{\alpha+1} e^{-L/L^*}\,\mathrm{d}\log L, \qquad \bar n_{2D} = n_{3D}\,\chi^2\Delta\chi. \tag{7}$$

Where a survey's catalogue count is published, the shot noise uses that count instead (Table 2). The survey bias always uses n_{3D} from the LF: incompleteness thins the sample at random and does not change its bias.

Survey bias. We use abundance matching: we find $M_{\rm min}$ with $n_{\rm ST}(>M_{\rm min}, z) = n_{3D}$ and set $b_{\rm survey} = b_{\rm SMT}(M_{\rm min}, z)$. Here $n_{\rm ST}$ is the Sheth & Tormen (1999) mass function, $b_{\rm SMT}$ is the Sheth, Mo & Tormen (2001) bias, and $\sigma(M)$ uses the Eisenstein & Hu (1998) transfer function. This is an upper value (no scatter, no duty cycle). For the validation we use La Plante's bias instead, $b_g(z) = 2.1(1+z) - 5.3$ (Waters et al. 2016, Sec. 5, Fig. 14).

Smallest usable multipole. A field of solid angle Ω cannot measure modes larger than itself, so

$$\ell_{\rm min} = \max\big(100,\ 2\pi/\sqrt{\Omega}\big) \qquad (\simeq 360 \text{ for } 1\,\mathrm{deg}^2). \tag{8}$$

Area for 3σ. At fixed depth and density, $(S/N)^2 \propto$ area, so $A_{3\sigma} = A\,(3/\mathrm{S/N})^2$.

2 Inputs and where every number comes from

2.1 CMB experiments

experiment	Δ_T [μK-arcmin]	FWHM	area [deg^2]	source
ACT DR6	9.6	$1.4'$	19,000	Naess+25 (2503.14451); beam $1.4'$ at 150 GHz
SPT-3G Main	1.9	$1.2'$	1,500	SPT-3G (2403.17925) Tab. 1; beam $1.2'$ assumed
SO	10	$1.4'$	16,000	SO science goals (1808.07445); post-ILC: LP22 Fig. 1
CMB-S4	2	$1.4'$	1,600	LP22 Tab. 1; post-ILC: LP22 Fig. 1; project ended 2025
CMB-HD	0.6	$10''$	20,000	LP22 Tab. 1; post-ILC: LP22 Fig. 1

Table 1: CMB experiments. Δ_T and the beam set the instrument-only noise. SO, CMB-S4 and CMB-HD also have La Plante's post-ILC curves (La Plante+22 Fig. 1, vector-digitised: `data/reference/la_plante_2022/fig1_filter_components_digitized.csv`). The area is the experiment's sky coverage; what enters $f_{\rm sky}$ is the overlap with the LAE field. CMB-S4: DOE and NSF ended support for the project in July 2025 (CASCA GAC report, Dec. 2025); it is kept here only for comparison.

CMB spectra.

- C_ℓ^{TT}: the lensed primary CMB of La Plante+22 Fig. 1 (digitised).
- $C_\ell^{\rm kSZ,reion}$: the seed-median auto-power of our own simulated kSZ map (reionisation era, $z \geq 6$). $D_{3000} = 0.95\,\mu{\rm K}^2$, against $1.15\,\mu{\rm K}^2$ in La Plante+22 Fig. 1.
- $C_\ell^{\rm kSZ,late}$: $D_\ell = 1.38\,(\ell/3000)^{0.21}\,\mu{\rm K}^2$ (La Plante+22 Eq. 9, from Park et al. 2018).

2.2 LAE surveys

survey	z	Δz	area	$L_{\rm min}$ [erg s^{-1}]	LF	$n(>L_{\rm min})$ [Mpc^{-3}]	N/deg^2 (LF)	N/deg^2 used	$b_{\rm ST}$
HSC SILVERRUSH	5.7	0.1	14	6.3×10^{42} (Weinberger+20 Tab. 1)	Umeda+25 Tab. 3 $z=5.7$	1.8×10^{-4}	174	69 (Ouchi+18)	4.8
HSC SILVERRUSH	6.6	0.1	21	7.9×10^{42} (Weinberger+20 Tab. 1)	Umeda+25 Tab. 3 $z=6.6$	3.2×10^{-5}	28	41 (Ouchi+18)	6.5
Subaru PFS	5.7	0.2	15	3.0×10^{42} (assumed)	Umeda+25 Tab. 3 $z=5.7$	6.5×10^{-4}	1256	1256 (LF)	4.1
Subaru PFS	6.6	0.2	15	3.0×10^{42} (assumed)	Umeda+25 Tab. 3 $z=6.6$	2.8×10^{-4}	493	493 (LF)	5.1
LAGER	6.9	0.076	10.19	6.3×10^{42} (Wold+22)	Wold+22 (4-field, $\alpha=-2.5$) $z=6.9$	7.1×10^{-5}	45	17 (Wold+22)	6.1
Roman deep (no LF evol.)	8	1	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	124	124 (LF)	7.8
Roman deep (no LF evol.)	9	1	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	112	112 (LF)	8.3
Roman deep (no LF evol.)	9.75	0.5	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	52	52 (LF)	8.7
Roman deep (LF as obs.)	8	1	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7.3$	1.5×10^{-6}	11	11 (LF)	9.6
Roman deep (LF as obs.)	9	1	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7.3$	1.5×10^{-6}	10	10 (LF)	10.2
Roman deep (LF as obs.)	9.75	0.5	1	1.0×10^{43} (Wold+23)	Umeda+25 Tab. 3 $z=7.3$	1.5×10^{-6}	4.8	4.8 (LF)	10.6
Roman 16 deg^2 (upper)	8	1	16	1.0×10^{43} (Wold+23 depth)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	124	124 (LF)	7.8
Roman 16 deg^2 (upper)	9	1	16	1.0×10^{43} (Wold+23 depth)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	112	112 (LF)	8.3
Roman 16 deg^2 (upper)	9.75	0.5	16	1.0×10^{43} (Wold+23 depth)	Umeda+25 Tab. 3 $z=7$	1.7×10^{-5}	52	52 (LF)	8.7
Roman HLWAS deep tier	8	1	19.2	4.9×10^{43} (ROTAC 2025, derived)	Umeda+25 Tab. 3 $z=7$	7.8×10^{-8}	0.58	0.58 (LF)	12.1
Roman HLWAS deep tier	9	1	19.2	6.4×10^{43} (ROTAC 2025, derived)	Umeda+25 Tab. 3 $z=7$	1.8×10^{-8}	0.12	0.12 (LF)	14.0
Roman HLWAS deep tier	9.75	0.5	19.2	7.6×10^{43} (ROTAC 2025, derived)	Umeda+25 Tab. 3 $z=7$	5.7×10^{-9}	0.018	0.018 (LF)	15.5

Table 2: LAE survey shells. $L_{\rm min}$: SILVERRUSH from Weinberger, Kulkarni & Haehnelt (2020) Table 1; LAGER from the faintest LAEs of Wold et al. (2022); Roman from the line-flux limits of Wold et al. (2023), who recover $L > 10^{43}\,{\rm erg\,s^{-1}}$ at 50% completeness at $z = 7.5$–10.5; PFS is an assumption. LFs: Umeda et al. (2025) Table 3 Schechter fits; LAGER four-field fit of Wold et al. (2022) with $\alpha = -2.5$. Observed counts: SILVERRUSH homogeneous samples of Ouchi et al. (2018), 959 LAEs in 13.8 deg^2 ($z = 5.7$) and 873 in 21.2 deg^2 ($z = 6.6$); LAGER 174 candidates in 10.19 deg^2 (Wold et al. 2022). Areas: SILVERRUSH 14/21 deg^2 (Konno et al. 2018); LAGER 3.24+2.91+1.90+2.14 deg^2 (WIDE12, GAMA15A, COSMOS, CDFS; Wold et al. 2022); Roman deep field 1 deg^2 (Wold et al. 2023). Roman 16 deg^2: the area of Maitra et al. (2025) Table 1 at the deep-field depth (upper estimate). Roman HLWAS deep tier: 19.2 deg^2 (ROTAC report, 2025), grism limit $\sim 6\times10^{-17}\,{\rm erg\,s^{-1}\,cm^{-2}}$ (our derivation from the report: 2.5× deeper than the medium tier's 1.5×10^{-16}), $L_{\rm min} = 4\pi d_L^2 F$ at the shell centres. Roman: two LF choices bracket the density, no evolution from $z = 7$ (optimistic) and the $z = 7.3$ LF (the observed decline). $b_{\rm ST}$: abundance-matched bias at the shell redshift.

Comparison with Maitra et al. (2025). Their Table 1 lists the LAE surveys used in their kSZ×LAE forecast. Where our numbers differ: SILVERRUSH ∼25 deg^2 (all narrow bands together) against our 21 deg^2 for the $z = 6.6$ band alone; LAGER ∼3 deg^2 (one field) against our 10.19 deg^2 (the four-field sample of Wold et al. 2022); PFS 15 deg^2, as ours; Roman grism ∼16 deg^2 against our

$1\,\mathrm{deg}^2$ deep field. For Roman we keep the $1\,\mathrm{deg}^2$ deep field (Girish's "deep Roman survey") and add two wider variants:

- **Roman $16\,\mathrm{deg}^2$ (upper):** Maitra's area at the deep-field depth ($L > 10^{43}\,\mathrm{erg\,s^{-1}}$, $z = 7$ LF). Upper estimate.
- **Roman HLWAS deep tier:** the closest real Roman survey in area. Yung et al. (2023) have no $\sim$16 deg^2 tier (their HLS is $1700\,\mathrm{deg}^2$, with a $2.5\,\mathrm{deg}^2$ moderate-depth and a $0.5\,\mathrm{deg}^2$ ultra-deep field); the HLWAS deep tier of the Roman Core Community Surveys (ROTAC report) is $19.2\,\mathrm{deg}^2$, but its grism reaches only $L_{\min} \approx 5$–$8 \times 10^{43}\,\mathrm{erg\,s^{-1}}$ at $z = 8$–9.75, so it holds < 1 LAE per deg^2 per shell.

Footprint overlap. The SILVERRUSH (HSC-SSP Deep/UltraDeep), PFS and LAGER fields all lie at Dec $> -30°$. They are inside the ACT DR6 and SO footprints, except the HSC ELAIS-N1 field (Dec $+54°$), which is not removed here. They are all outside SPT-3G Main (Dec $\approx -42°$ to $-70°$), so SPT-3G × SILVERRUSH has zero overlap; we show it as a hypothetical. The position of the Roman deep field is not fixed; we assume it lies inside the CMB footprint.

2.3 Simulated signal

- **Simulation.** py21cmfast v4 (Mesinger et al. 2011; Davies et al. 2025): 300 cMpc boxes, 300^3 (velocity, ionisation) and 600^3 (density, haloes) cells, $z = 5$–20, ten seeds, the 21cmFAST default cosmology ($H_0 = 67.77$, $\Omega_m = 0.3086$, $\Omega_b = 0.0489$). Velocities are converted to peculiar velocities, $v_{\mathrm{pec}} = v_z/(1+z)$, and the density is scaled to z with the linear growth factor.
- **Maps.** Wrap-cycle lightcones and a reionisation-era kSZ map ($z \geq 6$). For each seed and each $\Delta z = 1$ window ($z_0 = 6.5$–13 in steps of 0.5) we build a galaxy map: either matter × the Waters+16 bias (La Plante's model) or the projected overdensity of a simulated tracer (haloes; LAEs from Jahaan's Lyα luminosity catalogue built on the same haloes; LBGs from M_{UV}).
- **Spectra.** We filter the kSZ map with each experiment's f, square it, and measure $C_\ell^{\mathrm{kSZ}^2\times g}$ and C_ℓ^{gg} in 40 log bins, $\ell = k\chi_{\mathrm{eff}}$ (script 53).
- **Seed scatter.** $C_\ell^{\mathrm{kSZ}^2\times g}$ is the mean of ten seeds of a $\sim 4\,\mathrm{deg}^2$ patch, so we use $\left(C_\ell^{\mathrm{kSZ}^2\times g}\right)^2 \to \max(\langle C\rangle^2 - \sigma^2_{\langle C\rangle}, 0)$, which removes the seed scatter.
- **Rescaling to a survey.** With $r = b_{\mathrm{survey}}/b_{\mathrm{sim}}(z)$ (b_{sim}: the large-scale bias measured in the simulation, $\sum P_{tm}/\sum P_{mm}$ at $k < 0.15\,\mathrm{Mpc}^{-1}$; Waters+16 for the matter run):

$$C_\ell^{\mathrm{kSZ}^2\times g} \to r\, C_\ell^{\mathrm{kSZ}^2\times g}, \qquad C_\ell^{gg} \to r^2\big(C_{\ell,\mathrm{sim}}^{gg} - C_{\mathrm{sim}}^{\mathrm{SN}}\big)\frac{\Delta\chi_{\mathrm{win}}}{\Delta\chi_{\mathrm{shell}}} + \frac{1}{\bar{n}_{2D}}. \tag{9}$$

  The factor $\Delta\chi_{\mathrm{win}}/\Delta\chi_{\mathrm{shell}}$ is the Limber scaling of the clustering with shell depth (narrow-band shells are $\Delta z \approx 0.1$).
- **Two signals.** "La Plante's galaxy model" (the matter run) is the validated setup (Sec. 3); it may be low by up to $\sim$2×. "Simulated LAEs" ($L_{\mathrm{Ly}\alpha} \geq 10^{42.5}\,\mathrm{erg\,s^{-1}}$) give a 2–6× larger response per unit bias, as do the brightest LBGs and the $10^{12}\,M_\odot$ haloes. This excess is not yet confirmed (null test pending).

2.4 Digitised data from La Plante et al. (2022)

All files are in `data/reference/la_plante_2022/`; the README there describes each extraction.

- **Fig. 1** (filter ingredients, post-ILC noise): vector paths.
- **Fig. 8** (shot noise): the Roman HLS densities $N = 12{,}900$, 6200, 2040 and 340 galaxies per deg^2 per unit z at $z_0 = 7$–10, from W16.

- **Fig. 10** (cumulative S/N): vector paths; the end points reproduce Table 2.
- **Fig. 11** (per-bin errors for CMB-HD): pixel extraction.

3 Validation against La Plante et al. (2022)

We use exactly their setup: the Roman HLS area of $2200\,\mathrm{deg}^2$ ($f_{\mathrm{sky}} = 0.053$), their galaxy model (matter $\times$ W16 bias), their Fig. 8 shot noise, $\ell \geq 100$, and the single redshift bin $(z_0, \Delta z)$ that maximises the S/N, floored at $z = 6$ (their Table 2 and Fig. 9). Wide bins are built from consecutive $\Delta z = 1$ windows. The wide-bin galaxy map is the mean of the unit maps, so $C^{\mathrm{kSZ}^2\times g}$ is the mean of the unit C's and, in the Limber approximation, $C^{gg} = \sum_i C_i^{gg}/N^2$ (script 57).

CMB noise	shot noise	SO		CMB-S4		CMB-HD	
		LP22	ours	LP22	ours	LP22	ours
post-ILC	with	1.4	1	2.8	2.1	13	9.7
post-ILC	without	2.4	1	4.8	2.1	23	10
instrument (white)	with	1.7	1.3	11	9.4	35	24
instrument (white)	without	2.9	1.3	18	9.7	64	25

Table 3: Total S/N (best single redshift bin): La Plante+22 Table 2 (fiducial history) vs ours.

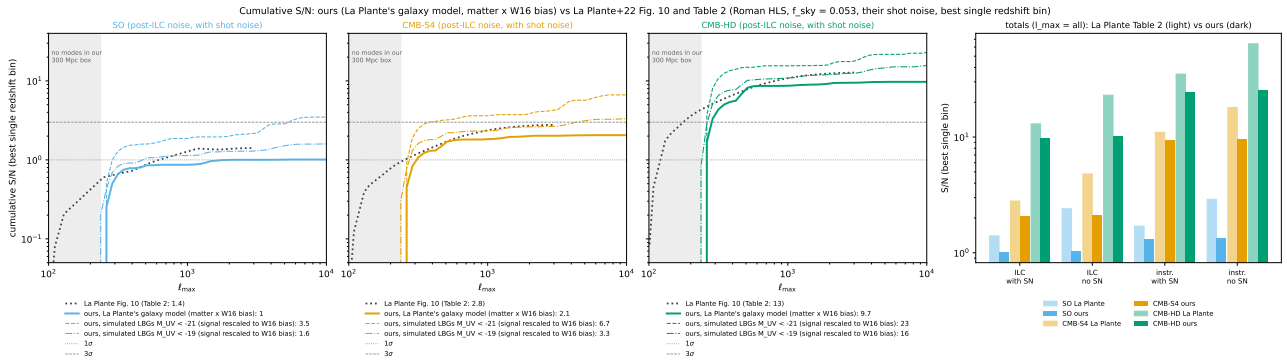


Figure 1: Cumulative S/N vs ℓ_{max} for SO, CMB-S4 and CMB-HD (post-ILC noise, with shot noise). Solid: ours (matter $\times$ W16 bias); dotted: La Plante+22 Fig. 10; dashed and dot-dashed: our simulated LBGs ($M_{\mathrm{UV}} < -21$, < -19), with the signal rescaled to the W16 bias. Grey band: no modes in our 300 Mpc box. Right panel: totals vs Table 2.

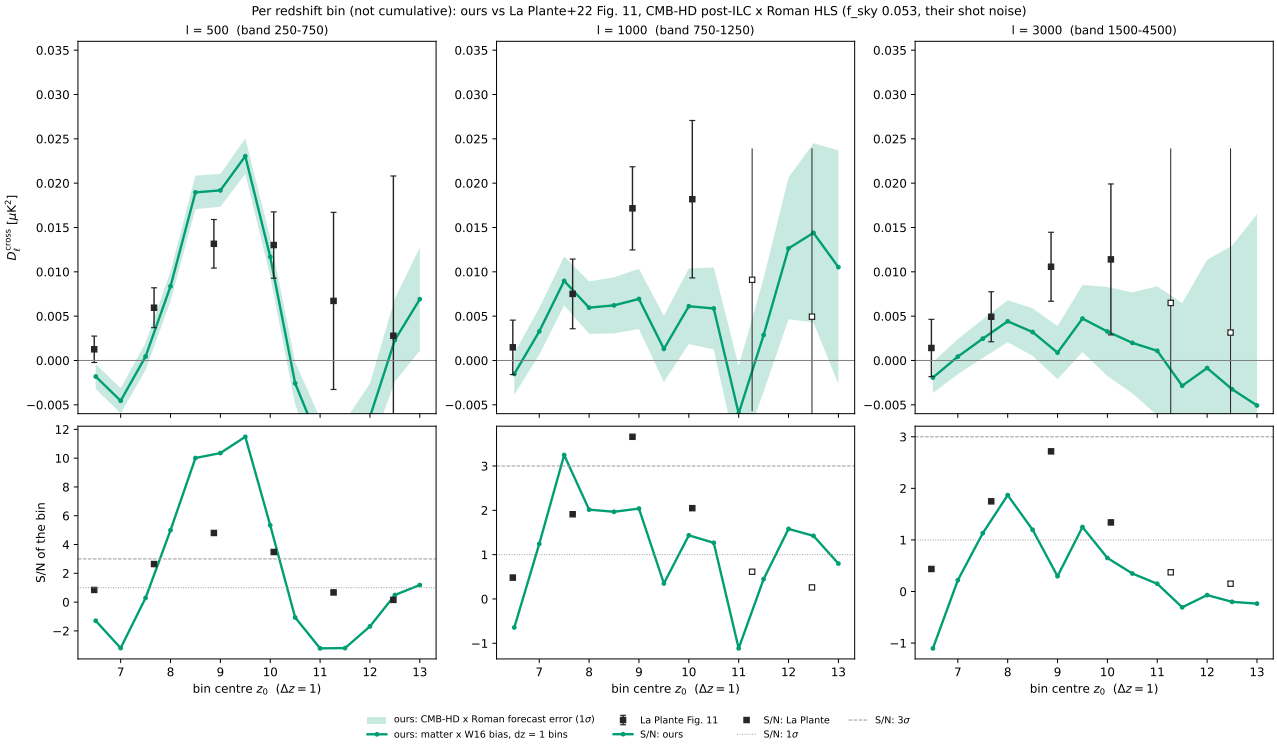


Figure 2: Per redshift bin ($\Delta z = 1$), CMB-HD post-ILC × Roman HLS: our D_ℓ with forecast errors (green) vs La Plante+22 Fig. 11 (black; open squares: error bar runs off their axis), in their three ℓ bands. Bottom row: S/N of each bin.

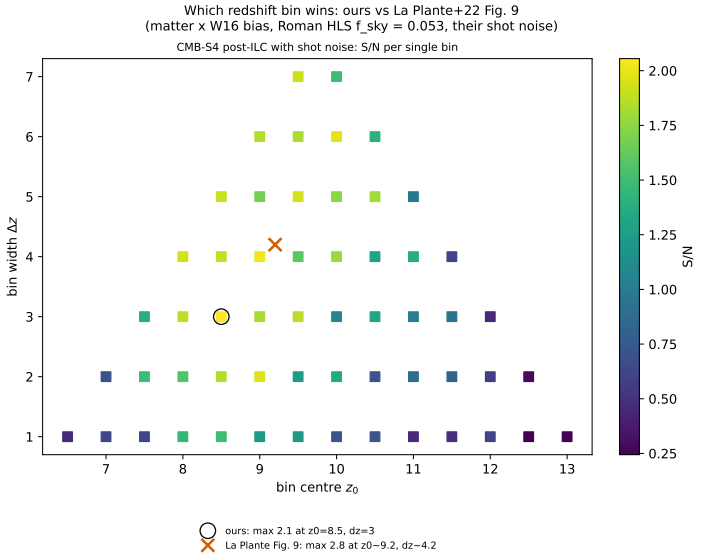


Figure 3: S/N of every single redshift bin $(z_0, \Delta z)$ for CMB-S4 (post-ILC, with shot noise), the analogue of La Plante+22 Fig. 9. Circle: our best bin; cross: theirs.

Findings.

- **Clustering.** Our galaxy clustering matches La Plante+22 Fig. 8 to ∼5% ($D^{gg}_{1000} = 0.16$, 0.21, 0.26, 0.28 at $z_0 = 7$–10).
- **Noise.** Our per-bin errors at $\ell \approx 1000$ and 3000 match their Fig. 11, so the noise side of the pipeline (filter, CMB noise, shot noise, sample variance) is validated.
- **Best bin.** The best single bin lies in the same region (ours $z_0 = 8.5$, $\Delta z = 3$; theirs $z_0 \approx 9.2$, $\Delta z \approx 4.2$).

- **Large scales.** Our box has no modes below $\ell \approx 240$. In their Fig. 10, for CMB-HD, 4.3 of the 12.8 total S/N comes from below that ℓ.
- **Signal amplitude.** Our signal at $\ell \gtrsim 1000$ is ∼0.4–0.5 of theirs near the peak. Likely causes are the box size (missing large-scale velocity modes) and our weaker simulated kSZ; the reionisation histories are similar ($x_{\rm HII} \approx 0.19$ vs 0.20 at $z = 9$). Our $\ell \approx 500$ band is unreliable: it contains only ∼2 Fourier modes of a 300 Mpc patch.
- **Open.** Their "without shot noise" totals are far above their "with" totals, while ours barely change; this is not understood yet.

4 Detectability for the requested CMB×LAE pairs

The pairs are: (1) existing data, ACT DR6 × HSC SILVERRUSH ($z \approx 6.6$); (2) a better CMB with the same LAEs, SO LAT × SILVERRUSH and SO LAT × LAGER ($z \approx 6.9$); (3) better LAEs, SO LAT × a deep Roman grism field ($z \approx 7.5$–10); (4) both better, CMB-HD × Roman deep. Pairs 3w and 4w repeat 3 and 4 with the wider Roman variants (Maitra et al. 2025 area; Sec. 2). We also show SPT-3G × SILVERRUSH as a hypothetical (no real overlap). Scripts 58 and 59.

pair	CMB (noise)	LAE survey	LAEs/deg^2	S/N, LP22 galaxy model	S/N, sim. LAEs	area for 3σ [deg^2]
1	ACT DR6 (white noise)	HSC SILVERRUSH	41	0.0074	0.0159	> full sky
1s	SPT-3G Main (white noise)	HSC SILVERRUSH †	41	0.0619	0.1066	> full sky
2a	SO (post-ILC)	HSC SILVERRUSH	41	0.0054	0.0118	> full sky
2a	SO (white noise)	HSC SILVERRUSH	41	0.007	0.015	> full sky
2b	SO (post-ILC)	LAGER	17	0.0031	0.0112	> full sky
2b	SO (white noise)	LAGER	17	0.004	0.0143	> full sky
3	SO (post-ILC)	Roman deep (no LF evol.)	124 / 112 / 52	0.0068	0.0659	> full sky
3	SO (white noise)	Roman deep (no LF evol.)	124 / 112 / 52	0.0088	0.0855	> full sky
3	SO (post-ILC)	Roman deep (LF as obs.)	11 / 10 / 4.8	0.0029	0.0267	> full sky
3	SO (white noise)	Roman deep (LF as obs.)	11 / 10 / 4.8	0.0037	0.0342	> full sky
4	CMB-HD (post-ILC)	Roman deep (no LF evol.)	124 / 112 / 52	0.0833	0.4678	1.3e+03
4	CMB-HD (white noise)	Roman deep (no LF evol.)	124 / 112 / 52	0.2156	1.171	194
4	CMB-HD (post-ILC)	Roman deep (LF as obs.)	11 / 10 / 4.8	0.0361	0.2179	6.9e+03
4	CMB-HD (white noise)	Roman deep (LF as obs.)	11 / 10 / 4.8	0.0933	0.5817	1.03e+03
3w	SO (post-ILC)	Roman 16 deg^2 (upper)	124 / 112 / 52	0.0383	0.2931	> full sky
3w	SO (white noise)	Roman 16 deg^2 (upper)	124 / 112 / 52	0.0498	0.3822	> full sky
3w	SO (post-ILC)	Roman HLWAS deep tier	0.58 / 0.12 / 0.018	0.0042	0.0478	> full sky
3w	SO (white noise)	Roman HLWAS deep tier	0.58 / 0.12 / 0.018	0.0056	0.0624	> full sky
4w	CMB-HD (post-ILC)	Roman 16 deg^2 (upper)	124 / 112 / 52	0.3997	1.9994	901
4w	CMB-HD (white noise)	Roman 16 deg^2 (upper)	124 / 112 / 52	1.0162	4.9328	139
4w	CMB-HD (post-ILC)	Roman HLWAS deep tier	0.58 / 0.12 / 0.018	0.0426	0.4431	> full sky
4w	CMB-HD (white noise)	Roman HLWAS deep tier	0.58 / 0.12 / 0.018	0.1099	1.1311	1.43e+04

† SPT-3G Main and SILVERRUSH do not overlap on the sky; computed as if they did (real S/N = 0).

Table 4: S/N for each pair (all ℓ, all shells of the survey; seed scatter removed). The LAEs/deg^2 column gives the value per shell. The 3σ area assumes the same depth and density.

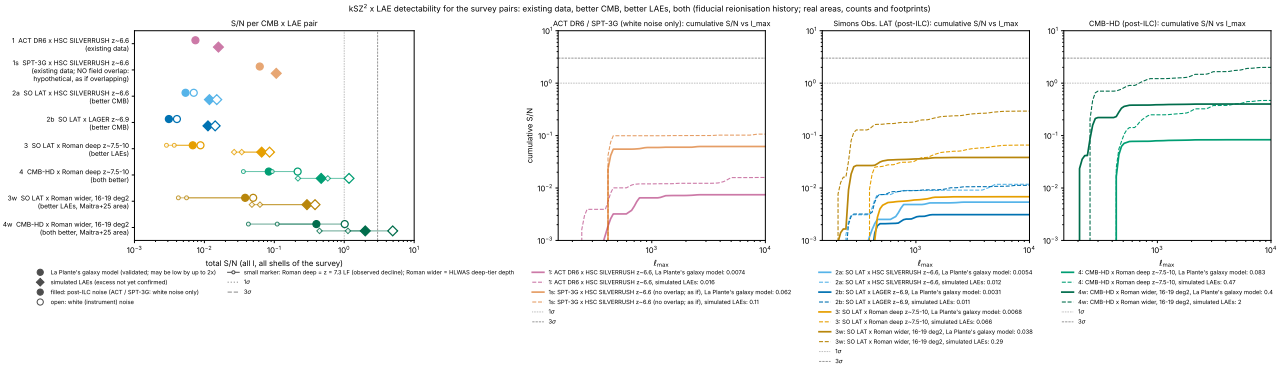


Figure 4: Left: total S/N per pair. Circles: La Plante's galaxy model (validated); diamonds: simulated LAEs. Filled: post-ILC noise (ACT, SPT-3G: white noise only); open: white noise. Small markers: Roman deep with the $z = 7.3$ LF; Roman wider at the HLWAS deep-tier depth (the large markers are the 16 deg^2 upper estimate). Right: cumulative S/N vs $\ell_{\rm max}$, one panel per CMB experiment.

Per-pair curves. Figures 5–12 show one pair each. Left: the S/N if the survey (same area, density, depth and shell width) sat at each $\Delta z = 1$ window; stars mark its real shells. Right: cumulative S/N for the real shells. Solid: simulated LAEs; dashed: La Plante's galaxy model; thick: post-ILC; thin: white noise. Grey band: the LAE bias is held constant beyond the measured redshift.

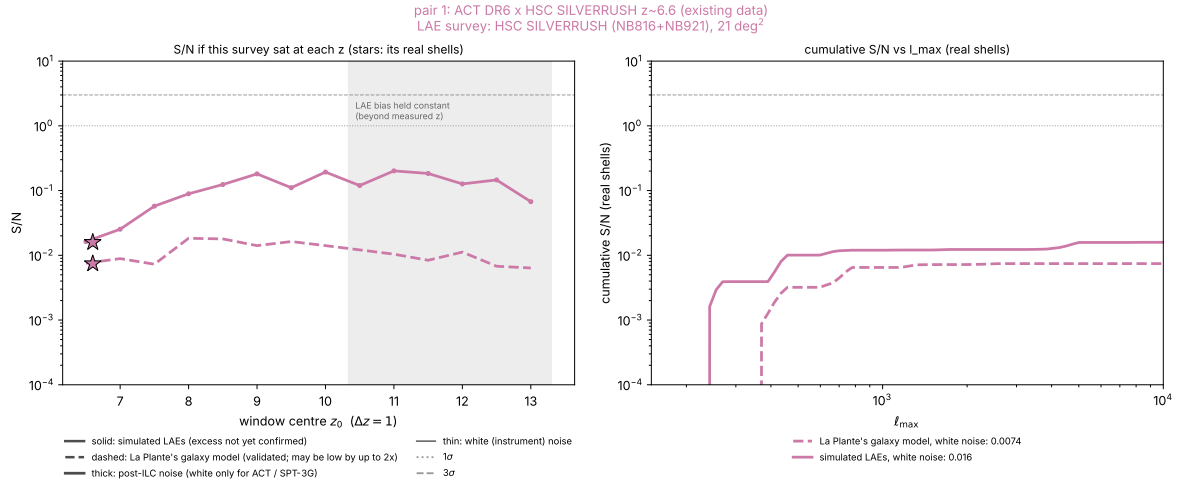


Figure 5: Pair 1: ACT DR6 × HSC SILVERRUSH ($z \approx 6.6$), existing data.

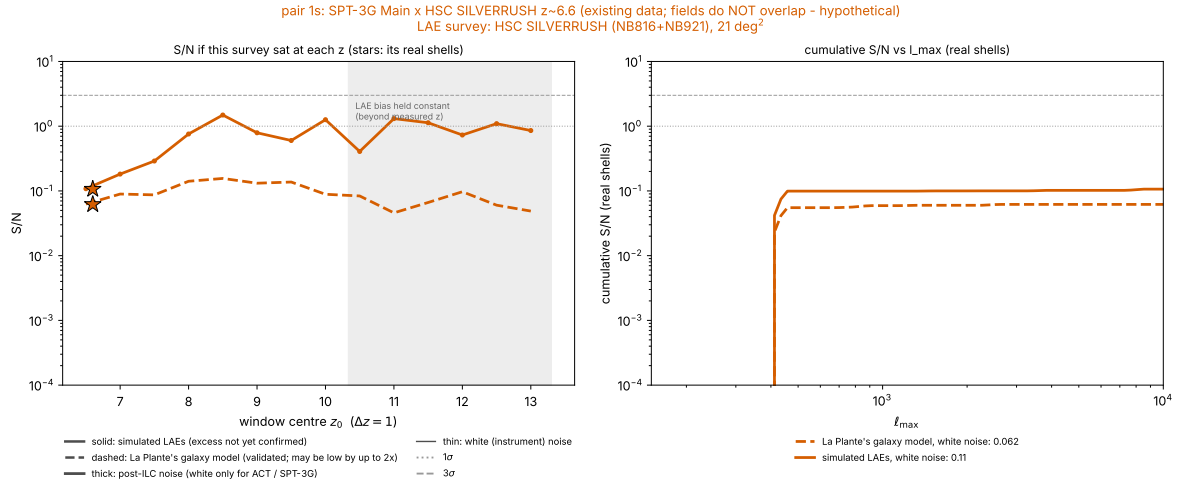


Figure 6: Pair 1s: SPT-3G Main × HSC SILVERRUSH, hypothetical (the fields do not overlap).

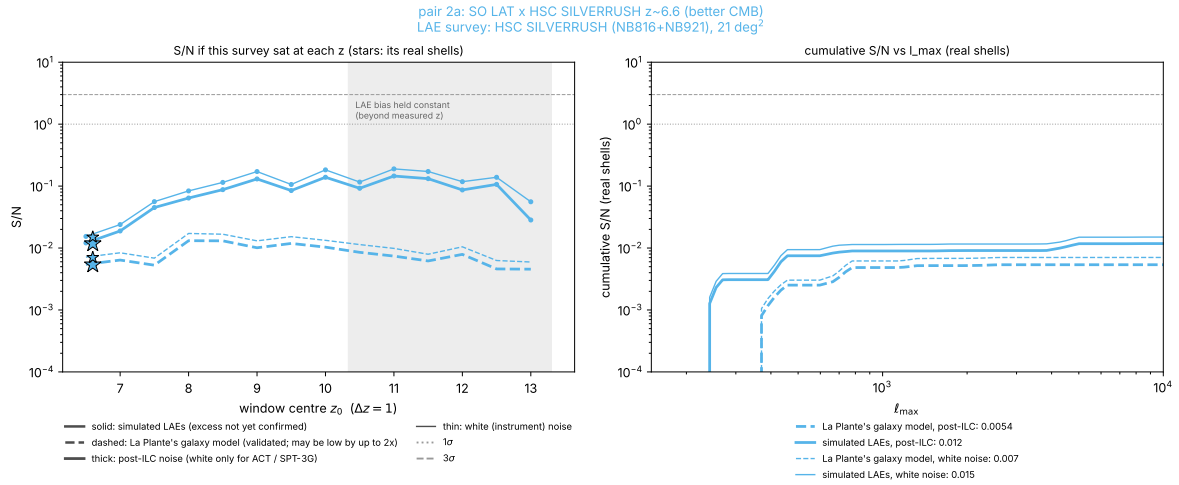


Figure 7: Pair 2a: SO LAT × HSC SILVERRUSH ($z \approx 6.6$), better CMB.

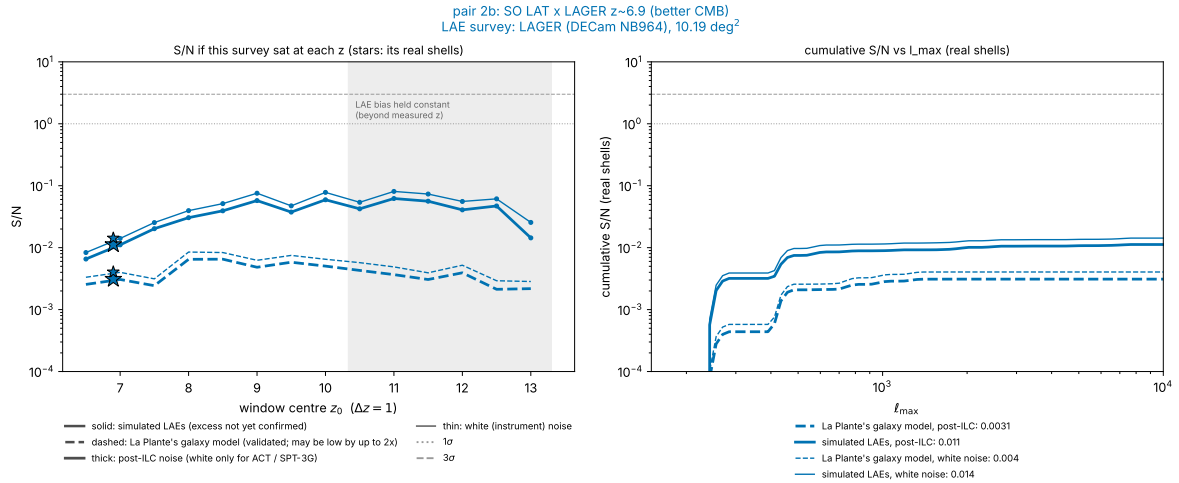


Figure 8: Pair 2b: SO LAT × LAGER ($z \approx 6.9$), better CMB.

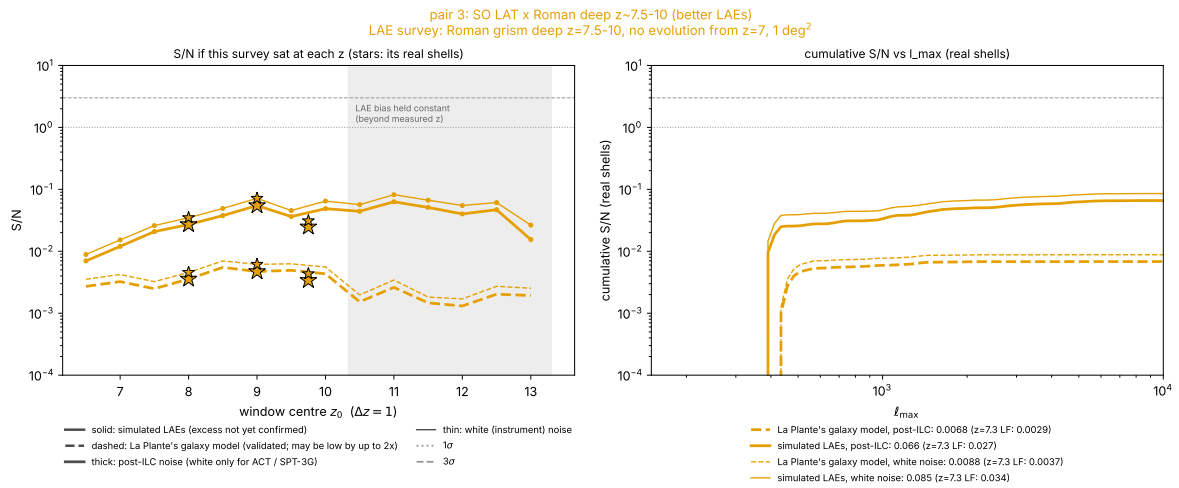


Figure 9: Pair 3: SO LAT × Roman deep ($z \approx 7.5$–10), better LAEs.

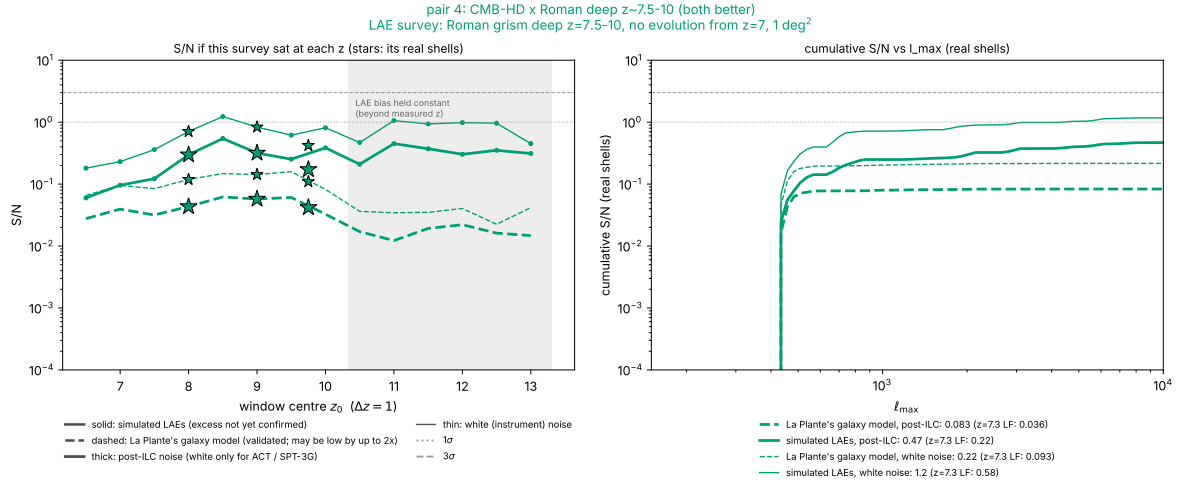


Figure 10: Pair 4: CMB-HD × Roman deep ($z \approx 7.5$–10), both better.

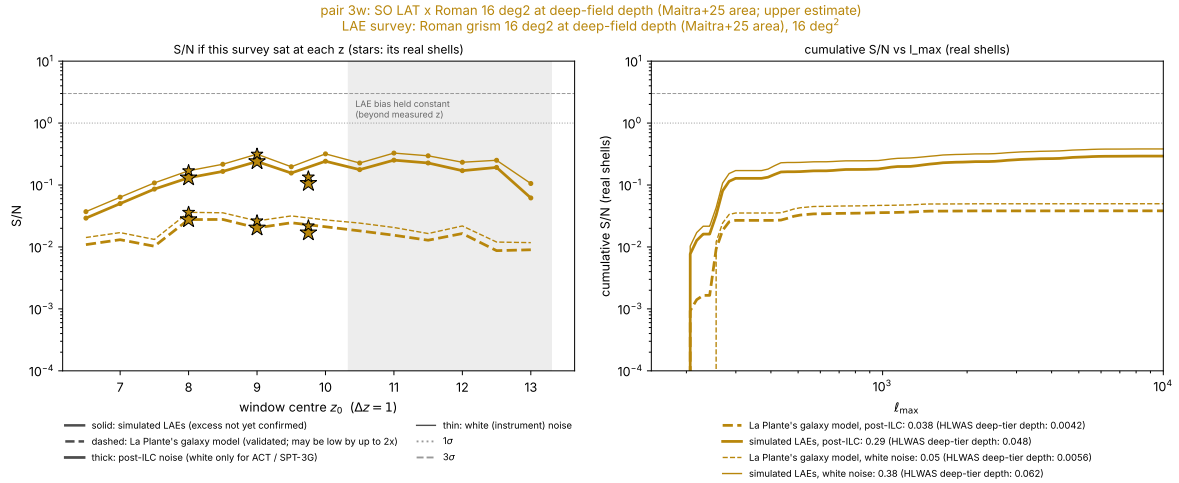


Figure 11: Pair 3w: SO LAT × Roman 16 deg^2 at deep-field depth (Maitra et al. 2025 area; upper estimate). The legend also gives the HLWAS deep-tier value.

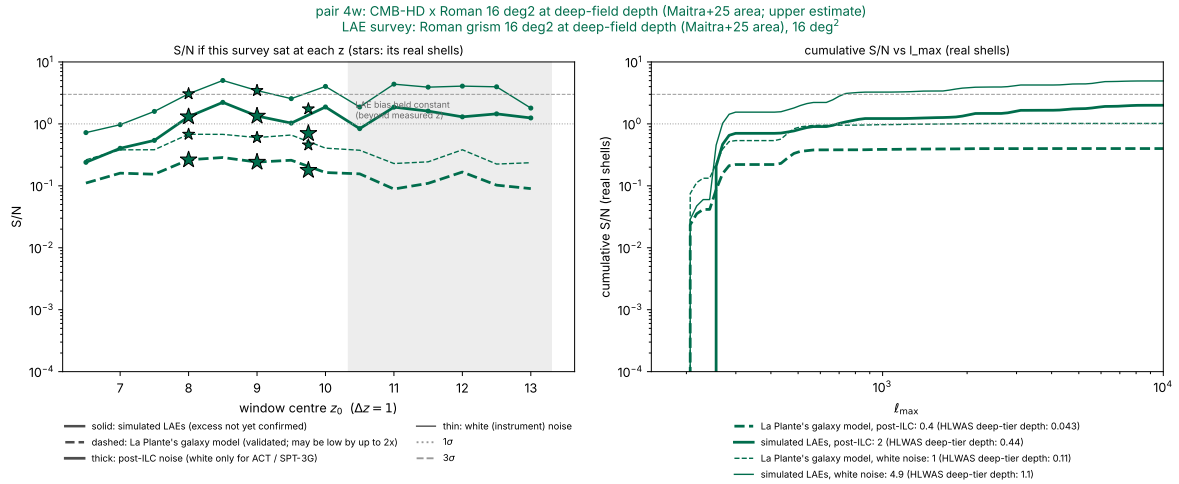


Figure 12: Pair 4w: CMB-HD × Roman 16 deg^2 at deep-field depth (Maitra et al. 2025 area; upper estimate). The legend also gives the HLWAS deep-tier value.

Reading.

- **Existing data** give S/N ∼0.01.
- **SO LAT** has a map depth similar to ACT DR6 (10 vs 9.6 μK-arcmin), and its larger sky area does not help a 10–20 deg^2 LAE field.
- **SPT-3G**'s much deeper map would give ∼8× ACT on the same field, if an LAE field existed inside its footprint.
- **Redshift.** The $z \approx 6.6$–6.9 surveys sit where the signal has mostly faded; the signal peaks at $z \approx 8$–9, where Roman observes.
- **Area.** Only CMB-HD × Roman approaches 1σ. The limit is the LAE survey area.
- **Wider Roman.** Going from 1 to 16 deg^2 at the same depth raises the S/N by ∼4–5 (CMB-HD: 0.08 to 0.4 post-ILC, 0.2 to 1.0 white, validated signal model). This needs a 16 deg^2 grism survey as deep as the 1 deg^2 field, which is not planned. At the real HLWAS deep-tier depth the gain in area is lost to the shallower grism (0.04).

5 All surveys × all CMB experiments

For completeness (CMB-S4 kept for comparison only), script 55:

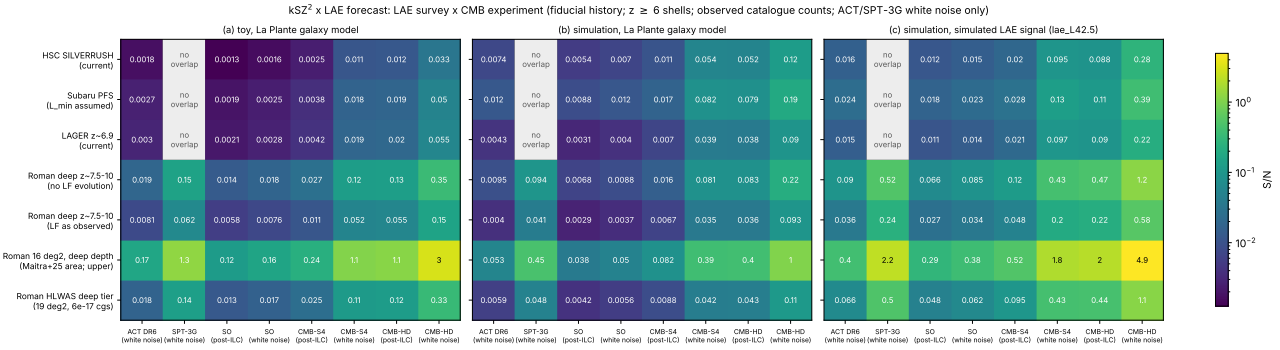


Figure 13: S/N for every LAE survey (rows) and CMB experiment (columns). (a) the toy model calibrated to La Plante+22 Table 2 (one rescaling factor, 1.45); (b) the simulation with La Plante's galaxy model; (c) the simulation with the simulated LAEs. Only shells at $z \geq 6$ are used (the simulation starts at $z = 5$, and the kSZ map at $z = 6$).

survey / signal	ACT DR6 (white noise)	SPT-3G (white noise)	SO (post-ILC)	SO (white noise)	CMB-S4 (post-ILC)	CMB-S4 (white noise)	CMB-HD (post-ILC)	CMB-HD (white noise)
HSC SILVERRUSH, LP22 model	0.0074	–	0.0054	0.007	0.011	0.054	0.052	0.12
HSC SILVERRUSH, sim. LAEs	0.016	–	0.012	0.015	0.02	0.095	0.088	0.28
Subaru PFS, LP22 model	0.012	–	0.0088	0.012	0.017	0.082	0.079	0.19
Subaru PFS, sim. LAEs	0.024	–	0.018	0.023	0.029	0.13	0.11	0.39
LAGER, LP22 model	0.0043	–	0.0031	0.004	0.007	0.039	0.038	0.09
LAGER, sim. LAEs	0.015	–	0.011	0.014	0.021	0.097	0.09	0.22
Roman deep (no LF evol.), LP22 model	0.0095	0.094	0.0068	0.0088	0.016	0.081	0.083	0.22
Roman deep (no LF evol.), sim. LAEs	0.09	0.52	0.066	0.086	0.12	0.43	0.47	1.2
Roman deep (LF as obs.), LP22 model	0.004	0.041	0.0029	0.0037	0.0067	0.035	0.036	0.093
Roman deep (LF as obs.), sim. LAEs	0.036	0.24	0.027	0.034	0.048	0.2	0.22	0.58
Roman 16 deg^2 (upper), LP22 model	0.053	0.45	0.038	0.05	0.082	0.39	0.4	1
Roman 16 deg^2 (upper), sim. LAEs	0.4	2.2	0.29	0.38	0.52	1.8	2	4.9
Roman HLWAS deep tier, LP22 model	0.0059	0.048	0.0042	0.0056	0.0088	0.042	0.043	0.11
Roman HLWAS deep tier, sim. LAEs	0.066	0.5	0.048	0.062	0.095	0.43	0.44	1.1

Table 5: The numbers of Fig. 13(b, c). "–": no footprint overlap.

6 Caveats and open items

- **Noise model.** Gaussian covariance as in La Plante+22. Super-sample variance, non-Gaussian kSZ^2 covariance and the real field geometry are not included; all of them would lower the S/N of the small LAE fields.

- **ACT and SPT-3G**: instrument noise only (no foreground residuals), so optimistic.
- **Signal amplitude.** Up to $\sim$2$\times$ below La Plante+22 at $\ell \gtrsim 1000$; a bigger box is needed (the planned large-box LAE runs with Jahaan).
- **Simulated-LAE excess.** Not yet confirmed. Next: a mismatched-seed null test and a halo sample matched in number density.
- **Survey bias.** Abundance matching without scatter, so an upper value. PFS $L_{\rm min}$ is assumed. The Roman density is bracketed by two LF choices. The HLWAS deep-tier grism limit is our derivation from the ROTAC report.
- **Footprints.** ELAIS-N1 is not removed from SILVERRUSH; the Roman deep-field position is assumed inside the CMB footprint.

Reproducing the numbers

Repository `ksz-lae-xcorr`. Script 53 (on the cluster) writes the binned spectra `data/products_tracers/sim_spectra/*.npz`; scripts 55, 57, 58 and 59 (on a desktop, seconds to minutes) write the figures and CSVs under `data/plots/2026-11-03_ksz_lae_snr/`; script 60 writes the tables of this report. The code is `src/ksz_lae_xcorr/snr/feasibility.py` (experiments, surveys, filter, bias) and `snr/sim_snr.py` (estimator on the simulated spectra).

References

- CASCA GAC report to the CASCA Board, Dec. 2025 (CMB-S4 status), casca.ca
- Davies, Mesinger, Murray et al. 2025, 21cmFAST v4, arXiv:2504.17254
- Eisenstein & Hu 1998, arXiv:astro-ph/9709112
- Ferraro et al. 2016 (projected-field kSZ estimator; via La Plante+22)
- Konno et al. 2018, SILVERRUSH IV, arXiv:1705.01222
- La Plante, Sipple & Lidz 2022, ApJ 928, 162, arXiv:2111.13717 (La Plante+22)
- Maitra et al. 2025, arXiv:2505.17188 (Table 1: LAE survey areas)
- Mesinger, Furlanetto & Cen 2011, 21cmFAST, arXiv:1003.3878
- Naess et al. 2025, ACT DR6 maps, arXiv:2503.14451
- Ouchi et al. 2018, SILVERRUSH I, arXiv:1704.07455
- Park et al. 2018 (late-time kSZ; via La Plante+22 Eq. 9), arXiv:1710.02792
- Roman Observations Time Allocation Committee (ROTAC) 2025, final report on the Core Community Surveys, arXiv:2505.10574
- Sheth & Tormen 1999, arXiv:astro-ph/9901122
- Sheth, Mo & Tormen 2001, MNRAS 323, 1, arXiv:astro-ph/9907024
- Simons Observatory collaboration 2019, science goals, arXiv:1808.07445
- SPT-3G collaboration 2024, arXiv:2403.17925
- Umeda et al. 2025, SILVERRUSH XIV, arXiv:2411.15495
- Waters et al. 2016, MNRAS 463, 3520, arXiv:1605.05670
- Weinberger, Kulkarni & Haehnelt 2020, arXiv:1911.11783
- Wold et al. 2022, LAGER Lyα LF at $z \sim 7$, arXiv:2105.12191
- Wold et al. 2023, Roman grism deep field, arXiv:2305.01562
- Yung et al. 2023, Roman ultra-deep field forecasts, arXiv:2210.04902